

Quasi-dynamic reconstruction of directed physiological networks from cross-sectional data using idopNetwork and path homology

Jiapeng Wang^{a,1}, Yu Wang^{b,c,1}, Xin Tang^{d,1}, Xinrui Luo^{a,c}, Huixian Yan^e, Wencheng An^e, Jiayi Li^g, Jiayu Xu^f, Rongling Wu^c, Huyile Liang^{a,c,*}

^a College of Mathematics Science, Inner Mongolia Normal University, Hohhot, 010022, China

^b Institute of Statistics and Big Data, Renmin University of China, Beijing, 100872, China

^c Beijing Key Laboratory of Topological Statistics and Applications for Complex Systems, Beijing Institute of Mathematical Sciences and Applications, Beijing, 101408, China

^d Internal Department, Beijing Renmin University Hospital, Beijing, 100872, China

^e Department of Endocrinology, Beijing Haidian Hospital, Beijing Haidian Section of Peking University Third Hospital, Beijing, China

^f First Medical Center of People's Liberation Army General Hospital, Beijing, China

^g Department of Gastroenterology, Beijing Chaoyang Hospital, Capital Medical University, Beijing, 100020, China

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ABSTRACT

Background: Excessive adiposity drives cardiometabolic morbidity through macroscopic systemic mechanisms rather than isolated metabolic defects. While conventional statistical models effectively estimate marginal biomarker associations, they cannot resolve the directed and cyclic regulatory dependencies underlying this pathogenesis. Network physiology offers a robust mathematical framework to delineate these complex, feedback-driven interactions.

Methods: Applying the idopNetwork framework to cross-sectional data from a primary clinical cohort ($N = 7,343$) and an external validation cohort ($N = 6,881$), we reconstructed directed, weighted hematometabolic networks across four body mass index (BMI) strata. This analytical pipeline mapped static clinical snapshots onto a continuous coordinate space to extract quasi-dynamic autoregressive trajectories. GLMY path homology was subsequently integrated to quantify macroscopic network rigidity via higher-order topological features.

Results: Progressive adiposity manifested a graded topological reorganization along the allostatic load gradient, characterized by the functional polarity inversion of core physiological hubs. Uric acid initially centralized as a primary conduit of metabolic load, preceding the transition of hemoglobin from a homeostatic anchor into a predictive promoter of systemic inflammation. GLMY path homology defined advanced obesity as a rigid systemic gridlock, quantified by the abnormal accumulation of persistent one-dimensional cyclic loops (β_1) and two-dimensional voids (β_2). This terminal architecture exhibited profound sexual dimorphism. Male adiposity degraded into hyper-reactive inflammatory β_1 cyclic tangling, whereas the female network experienced a near-complete depletion of preexisting β_2 voids upon entering advanced obesity, culminating in extensive topological barrenness and the attenuation of compensatory feedback.

Conclusions: The internal physiological ecosystem structurally deteriorates from robust homeostatic buffering into an advanced, sexually dimorphic allostatic deadlock. By decoupling directed regulatory flows from systemic confounding, this macroscopic algebraic approach challenges universal treatment protocols, highlighting the necessity for sex-specific precision interventions in managing obesity-driven cardiometabolic risk.

1. Background

Excessive adiposity drives cardiometabolic morbidity through

macroscopic systemic mechanisms rather than isolated metabolic defects [1–4]. From the perspective of network physiology, an increasing body mass index (BMI) correlates with concurrent dysregulation across

* Corresponding author. College of Mathematics Science, Inner Mongolia Normal University, Hohhot, 010022, China.

E-mail address: lianghuyile@imnu.edu.cn (H. Liang).

¹ These authors contributed equally.

multiple integrated systems, including the hematologic, hepato-renal, inflammatory, and hemodynamic axes [5]. Rather than occurring in isolation, these clinical shifts reflect a structural reorganization of internal regulatory networks adapting to chronic allostatic load [6]. Coordinated variations among leukocyte populations, erythrocyte parameters, and circulating metabolic indices represent macroscopic topological dependencies and systemic resource reallocation within a constrained physiological environment, without necessarily implying direct molecular signaling. The pathogenic trajectory of obesity is therefore shaped by these evolving multidimensional interactions, forming a hierarchical architecture that precedes overt homeostatic failure [7].

Conventional epidemiological analyses relying on multivariable regression or isolated biomarker tracking effectively estimate marginal associations but cannot delineate this overarching regulatory architecture [8,9]. While network models increasingly map these complex interactions [10–12], standard approaches such as Pearson correlation and Gaussian graphical models generate strictly undirected graphs, leaving the directionality and regulatory polarity of physiological dependencies unresolved [13]. Techniques offering directional inference, including Bayesian Networks [14] and Structural Equation Modeling [15], are frequently constrained by the assumption of a Directed Acyclic Graph (DAG) structure. This acyclic restriction explicitly precludes the identification of cyclic feedback loops, which operate as the fundamental mechanisms for biological buffering and allostatic homeostasis in real physiological environments [16]. Constructing hierarchical networks with signed, directed, and cyclic dependencies traditionally requires high-resolution longitudinal tracking or targeted experimental interventions [17,18], a level of temporal granularity rarely available in large-scale clinical cohorts. While pseudotemporal algorithms effectively extract dynamic trajectories from static single-cell transcriptomic snapshots [19], applying these techniques directly to macroscopic clinical records presents significant methodological challenges. Unlike cellular differentiation, which typically follows unidirectional branching lineages, macroscopic physiological variables continuously modulate one another through dense, cyclic allostatic feedback loops [20,21].

Addressing this primary analytical bottleneck requires a mathematical structure capable of inferring feedback-driven topologies strictly from cross-sectional datasets. The informative, dynamic, omnidirectional, and personalized matrix autoregressive (idop-MAR) framework achieves this by projecting static clinical snapshots onto a continuous, quasi-dynamic coordinate space [22–27]. Treating physiological variables as interactive components within a resource-constrained environment, the algorithm extracts multidimensional autoregressive interactions, yielding sparse, hierarchical networks characterized by signed and directional regulatory edges. Integrating this dynamic inference model with GLMY path homology further enables the rigorous evaluation of higher-order topological complexity [28,29]. Tracking persistent mathematical features—specifically one-dimensional cyclic loops (β_1) and two-dimensional topological voids (β_2)—facilitates the extraction of exact algebraic signatures associated with structural regulatory shifts and systemic allostatic collapse.

We directed this dynamic inference strategy toward human hematometabolic regulation, applying regularized estimation to independently infer networks and capture stratum-specific structural rewiring across progressive BMI stages [13,30]. Moving past isolated, single-variable marginal associations to directly map system-level topological dynamics [17,31], this approach fundamentally shifts the analytical baseline to pinpoint the structural transitions preceding irreversible allostatic deadlock. Rather than merely driving independent quantitative deviations across isolated clinical parameters [2,3], progressive adiposity is hypothesized to trigger a systemic topological reconfiguration, fundamentally altering the regulatory polarity of key physiological variables. Recognizing the inherent biological disparities in lipid distribution, endocrine capacity, and immune reactivity between sexes, we further postulate that this overarching network collapse lacks

physiological uniformity. The transition into advanced clinical obesity is anticipated to bifurcate into profoundly sexually dimorphic topological trajectories, thereby challenging universal cardiometabolic management strategies.

Translating these methodological insights into public health applications holds particular relevance for populations undergoing rapid metabolic transitions amidst environmental and resource constraints. By deriving quasi-dynamic trajectories from accessible, cross-sectional clinical snapshots rather than relying on resource-intensive longitudinal cohorts, the idopNetwork pipeline offers a highly scalable network-screening framework. Deploying this analytical approach provides healthcare systems with a rigorous theoretical basis to evaluate population-level cardiometabolic risks. Embedding this topological monitoring into broader regional development strategies establishes a critical foundation for societal resilience. Leveraging advanced mathematical modeling to map population health trajectories directly supports the construction of robust public health defense systems. This macroscopic perspective facilitates the translation of complex network features into actionable preventive care, seamlessly aligning precision medicine with macro-level public health imperatives.

Building on this dynamic inference framework, the present investigation was designed to reconstruct directed, signed, and weighted hematometabolic networks across four BMI strata from cross-sectional health-examination data; to quantify higher-order topological features of these networks via GLMY path homology; to characterize the structural reorganization of core physiological hubs—including the exhaustion dynamics of hematopoietic buffering (RDW-SD)—and its sexual dimorphism along the ALI gradient; and to assess the internal statistical stability of the inferred network features and their reproducibility in an independent national survey cohort.

2. Materials and methods

2.1. Study population and data representation

The primary analysis utilized cross-sectional data from 7343 individuals undergoing routine health assessments at the Hospital of Renmin University of China (RUC) between 2024 and 2025. We incorporated an independent cohort of 6881 participants from the Korea National Health and Nutrition Examination Survey (KNHANES) for external validation. Baseline demographic characteristics—specifically age and gender—and clinical parameters for both cohorts, stratified by body mass index (BMI), are summarized in Table 1 and Table S1.

We reconstructed directed, signed, and weighted cyclic regulatory networks using the idopNetwork framework. The computational input consists of a cross-sectional biomarker matrix $X \in \mathbb{R}^{n \times p}$, where n is the total number of subjects and p is the number of clinical variables. The primary derivation cohort included $p = 21$ distinct variables, covering routine hematologic, inflammatory, hepatic, renal, and cardiometabolic indices. To maintain cross-cohort comparability, the external validation network was restricted to a shared subset of nine core parameters ($p = 9$).

Prior to network inference, all continuous physiological variables were standardized to eliminate scaling artifacts. To map state-specific regulatory architectures, the cohorts were divided into four standard BMI strata: underweight, normal weight, overweight, and obesity. Networks were inferred independently for each subgroup. This stratification provides a consistent mathematical baseline, ensuring that identified topological variations reflect structural shifts in biological regulation rather than cross-sectional sampling variance.

2.2. Inclusion and exclusion criteria

To minimize confounding effects from acute physiological perturbations and avoid imputation artifacts, complete-case analyses were performed for both cohorts.

Table 1

Baseline characteristics of the primary study population (RUC). Data are presented as mean $\pm$ standard deviation (SD) for continuous variables and n (%) for categorical variables. Age, body mass index (BMI), and 21 clinical parameters utilized for subsequent idopNetwork modeling are summarized. P-values were calculated using one-way ANOVA for continuous variables and Pearson's Chi-square test for categorical variables across BMI strata.

Variable	Overall (N = 7343)	Underweight (n = 1159)	Normal weight (n = 4837)	Overweight (n = 1058)	Obesity (n = 289)	P-value
Sex						<0.001
Female, n (%)	4424 (60.2%)	867 (74.8%)	3132 (64.8%)	351 (33.2%)	74 (25.6%)	
Male, n (%)	2919 (39.8%)	292 (25.2%)	1705 (35.2%)	707 (66.8%)	215 (74.4%)	
Age (Mean $\pm$ SD)	21.23 $\pm$ 3.59	20.42 $\pm$ 2.81	21.26 $\pm$ 3.54	21.80 $\pm$ 4.20	21.86 $\pm$ 4.24	<0.001
BMI (Mean $\pm$ SD)	21.97 $\pm$ 3.80	17.37 $\pm$ 0.92	21.35 $\pm$ 1.73	26.83 $\pm$ 1.36	33.04 $\pm$ 2.58	<0.001
SBP (Mean $\pm$ SD)	112.63 $\pm$ 12.90	107.65 $\pm$ 10.95	111.41 $\pm$ 12.11	119.77 $\pm$ 12.85	126.77 $\pm$ 13.51	<0.001
DBP (Mean $\pm$ SD)	67.16 $\pm$ 8.61	66.80 $\pm$ 7.85	66.37 $\pm$ 8.35	69.35 $\pm$ 9.11	73.80 $\pm$ 9.65	<0.001
WBC (Mean $\pm$ SD)	6.30 $\pm$ 1.52	5.96 $\pm$ 1.53	6.22 $\pm$ 1.47	6.73 $\pm$ 1.44	7.44 $\pm$ 1.70	<0.001
Hb (Mean $\pm$ SD)	141.26 $\pm$ 15.28	138.16 $\pm$ 13.70	139.93 $\pm$ 15.20	147.98 $\pm$ 14.50	151.17 $\pm$ 14.70	<0.001
MCV (Mean $\pm$ SD)	90.45 $\pm$ 4.95	90.96 $\pm$ 5.28	90.48 $\pm$ 4.98	89.98 $\pm$ 4.46	89.51 $\pm$ 4.43	<0.001
PLT (Mean $\pm$ SD)	255.39 $\pm$ 54.41	248.11 $\pm$ 54.57	253.87 $\pm$ 53.70	264.74 $\pm$ 54.63	275.90 $\pm$ 55.82	<0.001
MPV (Mean $\pm$ SD)	10.10 $\pm$ 1.00	10.09 $\pm$ 0.97	10.12 $\pm$ 1.00	10.06 $\pm$ 0.99	10.02 $\pm$ 0.95	0.123
RDW-SD (Mean $\pm$ SD)	42.84 $\pm$ 2.84	43.26 $\pm$ 2.89	42.91 $\pm$ 2.85	42.27 $\pm$ 2.64	42.17 $\pm$ 2.82	<0.001
Lym# (Mean $\pm$ SD)	2.13 $\pm$ 0.56	2.06 $\pm$ 0.55	2.10 $\pm$ 0.55	2.24 $\pm$ 0.57	2.39 $\pm$ 0.61	<0.001
Neu# (Mean $\pm$ SD)	3.65 $\pm$ 1.23	3.42 $\pm$ 1.29	3.60 $\pm$ 1.21	3.91 $\pm$ 1.14	4.41 $\pm$ 1.27	<0.001
RBC (Mean $\pm$ SD)	4.77 $\pm$ 0.51	4.65 $\pm$ 0.47	4.72 $\pm$ 0.50	5.01 $\pm$ 0.48	5.15 $\pm$ 0.48	<0.001
Mo (Mean $\pm$ SD)	0.37 $\pm$ 0.12	0.34 $\pm$ 0.12	0.36 $\pm$ 0.11	0.40 $\pm$ 0.11	0.44 $\pm$ 0.12	<0.001
Eo# (Mean $\pm$ SD)	2.10 $\pm$ 1.72	1.99 $\pm$ 1.61	2.10 $\pm$ 1.71	2.17 $\pm$ 1.83	2.33 $\pm$ 1.84	0.008
Ba# (Mean $\pm$ SD)	0.51 $\pm$ 0.27	0.53 $\pm$ 0.27	0.50 $\pm$ 0.27	0.49 $\pm$ 0.26	0.51 $\pm$ 0.27	0.017
ALT (Mean $\pm$ SD)	19.19 $\pm$ 18.16	13.61 $\pm$ 7.76	17.11 $\pm$ 14.37	27.89 $\pm$ 24.09	44.57 $\pm$ 38.02	<0.001
TBIL (Mean $\pm$ SD)	16.88 $\pm$ 6.96	17.51 $\pm$ 6.97	16.77 $\pm$ 7.04	16.82 $\pm$ 6.72	16.42 $\pm$ 6.25	0.007
DBIL (Mean $\pm$ SD)	3.36 $\pm$ 1.27	3.56 $\pm$ 1.32	3.34 $\pm$ 1.27	3.29 $\pm$ 1.22	3.19 $\pm$ 1.12	<0.001
SG (Mean $\pm$ SD)	1.02 $\pm$ 0.01	1.02 $\pm$ 0.01	1.02 $\pm$ 0.01	1.02 $\pm$ 0.01	1.02 $\pm$ 0.01	0.063
pH (Mean $\pm$ SD)	6.01 $\pm$ 0.52	5.97 $\pm$ 0.51	6.02 $\pm$ 0.53	6.02 $\pm$ 0.54	5.94 $\pm$ 0.53	0.003
Cr (Mean $\pm$ SD)	67.84 $\pm$ 12.98	63.89 $\pm$ 10.83	67.00 $\pm$ 12.76	73.94 $\pm$ 13.31	75.52 $\pm$ 13.11	<0.001
Uric (Mean $\pm$ SD)	364.47 $\pm$ 95.01	320.25 $\pm$ 76.72	352.15 $\pm$ 83.81	437.06 $\pm$ 97.49	482.34 $\pm$ 99.57	<0.001

For the primary RUC dataset, the baseline population included individuals attending routine health evaluations between 2024 and 2025. Inclusion required participants to be adults (≥ 18 years) with complete anthropometric records for body mass index (BMI) calculation. Participants were excluded based on any of the following criteria: (i) missing data for any of the 21 evaluated hematometabolic variables; (ii) documented clinical history or laboratory results indicating acute infection, which could transiently alter basal inflammatory and hematological profiles; or (iii) missing demographic data (age and sex).

A consistent screening protocol was applied to the external KNHANES validation cohort. Inclusion required adult status (≥ 18 years) and valid BMI measurements. Participants were excluded if they lacked complete data across the nine core variables or exhibited clinical signs of acute inflammatory stress. Following these criteria, the final analytical samples comprised 7343 participants in the primary cohort and 6881 in the validation cohort.

2.3. Quasi-dynamic embedding of cross-sectional data

The cross-sectional design of large-scale clinical cohorts precludes the direct application of classical differential equations, which require longitudinal data. To address this limitation without assuming temporal causality, the idopNetwork maps static multidimensional observations onto a monotonic coordinate, enabling the quasi-dynamic estimation of regulatory interactions.

To construct this continuous manifold, the aggregate physiological burden for each participant i was quantified using the Allostatic Load Index (ALI_i):

$$ALI_i = \sum_{j=1}^p z_{ij}$$

where z_{ij} represents the standardized measurement of biomarker j for subject i across the p clinical variables. An unweighted summation was utilized to avoid predefined clinical weighting, capturing the global mean shift associated with hematometabolic dysregulation. Sorting the cohort in ascending order of ALI_i establishes a continuous gradient,

defined as the ALI axis. Instead of chronological time, this monotonic space serves as a quantitative proxy for cumulative physiological burden, defining the macroscopic state of the system.

Conceptually rooted in allometric scaling, this mathematical embedding translates macroscopic ecosystem dynamics to the human internal milieu. As systemic burden escalates, physiological subsystems continuously adjust their regulatory interactions across hematologic, inflammatory, and metabolic compartments due to shared resource constraints. Sequencing the cross-sectional cohort along the ALI axis maps the topological evolution of the physiological system. This transformation converts discrete clinical snapshots into continuous quasi-dynamic trajectories, facilitating subsequent differential modeling. Crucially, within this ALI -ordered manifold, each inferred network edge must be interpreted strictly as a directional predictive dependency conditional on aggregate physiological burden, rather than definitive chronological causality. Accordingly, references to terms such as 'transition', 'phase shift', 'phase transition', or 'progression' throughout this work explicitly denote movement along this continuous, cross-sectionally derived gradient, rather than actual temporal dynamics.

2.4. Power-law trajectory modeling and allometric scaling

Biological and macroecological networks often exhibit allometric scaling, where the functional output of specific components scales nonlinearly with aggregate system size or burden [32]. Based on this principle, the quasi-dynamic trajectory of each hematometabolic biomarker, denoted as $v_j(ALI)$, was parameterized against the continuous macroscopic axis (ALI) using a power-law function:

$$v_j(ALI) = \alpha_j \cdot ALI^{\beta_j}$$

Here, α_j represents the normalization constant, and β_j is the allometric scaling exponent. All continuous variables were first scaled to $[0, 1]$ by min-max normalization. To estimate the parameters, the model was linearized by taking natural logarithms: $\log y = \log a + b \cdot \log x$. Observations with $\tau > 0$ and $y > 0$ were retained to guarantee domain validity, and ordinary least-squares regression was applied to the log-log-transformed data, yielding b as the slope and $a = \exp(\text{intercept})$. The

goodness of fit of each biomarker-specific trajectory was assessed by the coefficient of determination (R^2), the root-mean-square error (RMSE) computed in the original power-law domain, and the regression p-value; complete diagnostics for all biomarkers across the four BMI strata are reported in [Supplementary Table S2](#). The fitted curves were then sampled at 1000 uniformly spaced points along the *ALI* axis to provide smooth, continuously differentiable trajectories that satisfy the continuity requirements of the subsequent differential modeling.

This power-law model quantifies the adaptive responses of distinct physiological subsystems to increasing allostatic load. The function acts as a non-linear low-pass filter, reducing baseline inter-individual heterogeneity and cross-sectional sampling noise. Projecting these parameterized curves across the *ALI* gradient converts discrete clinical data into continuously differentiable trajectories. These smooth, state-dependent functions fulfill the continuity requirements necessary for integrating ordinary differential equations in the subsequent modeling steps.

2.5. Quasi-dynamic differential modeling and network reconstruction

By substituting temporal derivatives with gradients computed along the continuous *ALI* manifold, we derived a quasi-dynamic system of ordinary differential equations (ODEs):

$$\frac{dv_j(ALI)}{dALI} = f_{jj}(v_j(ALI)) + \sum_{k=1, k \neq j}^p f_{jk}(v_k(ALI))$$

This differential structure separates the trajectory of biomarker j into an intrinsic self-regulatory component (f_{jj} , reflecting basal maintenance) and the interactive effects (f_{jk}) exerted by other physiological variables. To estimate these latent functions without imposing rigid parametric assumptions, we applied a non-parametric basis-expansion strategy. This approach models nonlinear biological interactions while ensuring a numerically stable reconstruction of the interaction topology.

Since physiological variables are rarely mutually connected across the entire system, the ODE estimation incorporates network sparsification to identify critical regulatory coefficients. To ensure cross-stratum comparability, all hyperparameters were fixed identically across the four BMI strata. Trajectory estimation employed a Legendre basis expansion of order 3 with cumulative integration along the *ALI* manifold, coupled with adaptive sparse-group lasso (ASGL) regularization ($\lambda_1 = 1 \times 10^{-4}$, $\alpha = 0.5$) under Bayesian information criterion (BIC) selection. To prioritize intrinsic homeostasis over cross-system noise, differential group weights penalized self-regulatory groups less strictly (0.8) relative to cross-variable groups (1.0), thereby preserving the autonomous dynamics of individual subsystems. Furthermore, an endpoint-deviation constraint enforced the terminal ratio of the self-effect to the total effect within $[0.7, 0.8] \cup [1.2, 1.3]$, with cross-variable effects bounded at 120% of the target magnitude. Following support shrinkage, cross-variable groups were retained only when their coefficient norm reached at least 50% of the significance threshold ($\theta_{min} = 0.2$) relative to the self-regulatory group norm.

After filtering spurious signals using this regularized estimation, the cumulative dynamic influence along the *ALI* manifold was quantified by integrating the estimated regulatory functions. These integral values populate the weighted, asymmetric adjacency matrix G . Each directed element g_{jk} represents the targeted regulatory impact of biomarker k on j , with the algebraic sign indicating activation or inhibition and the absolute magnitude reflecting interaction strength. From this matrix, we computed positive and negative out-degree centralities to identify dominant hub variables and map hierarchical shifts under escalating allostasis stress.

Finally, to compare this quasi-dynamic framework with established methods, three baseline network models were constructed across the respective BMI strata. These included undirected Spearman correlation networks to capture global marginal covariations, undirected Gaussian

Graphical Models (GGMs) estimated via the Graphical Lasso (L_1 regularization) [13] to infer sparse partial correlations, and directed Pseudo-Time-Lagged Cross-Correlation (TLCC) networks to approximate linear directed associations ([Fig. S1](#)).

2.6. Subsampling-based stability analysis

Subsampling-based stability analysis. For each BMI stratum, 80% of subjects were drawn without replacement ($B = 100$ replicates), and the network was re-estimated under identical hyperparameters in each replicate. The selection frequency of each directed edge across replicates (0–100%) was recorded and visualized as a heatmap ([Fig. S2](#)). Because sampling proceeded without replacement, this procedure corresponds to subsampling rather than classical bootstrap resampling and is framed within the stability-selection methodology [33].

2.7. Algebraic topological analysis via GLMY path homology

To quantify higher-order structural complexity within the inferred networks, we applied the Grigor'yan, Lin, Muranov, and Yau (GLMY) path homology framework [28,29]. Unlike classical simplicial homology, which cannot natively process directed graphs, GLMY path homology accommodates asymmetric architectures. This mathematical property enables the mapping of directed regulatory flows and cyclic feedback loops.

Continuous filtrations were generated by systematically thresholding the edge weights of the estimated adjacency matrices, allowing the assessment of multiscale topological features. Homology was computed over the integer field ($\mathbb{Z}$) to avoid numerical torsion. The analysis tracked the persistence of Betti numbers (β_k) to associate algebraic properties with biological network states. The framework quantified β_0 (connected components), β_1 (one-dimensional directed cycles), and β_2 (two-dimensional topological voids). Within the biological context of this study, persistent β_1 features represent sustained, cyclic physiological feedback loops, whereas the emergence of β_2 voids indicates localized pathway blockages or topological unreachability within the metabolic network.

Independent filtrations were performed on purely positive (promoting) subgraphs, purely negative (inhibiting) subgraphs, and the complete network. This stratification mathematically separates baseline compensatory adaptations from established pathogenic network structures.

2.8. External validation cohort and sub-network construction

Independent verification of the idopNetwork framework utilized an external cohort from the Korea National Health and Nutrition Examination Survey (KNHANES). Because this public database lacks certain micro-hematological parameters present in the primary dataset, we constructed a restricted nine-variable validation sub-network. This targeted model incorporated only the shared core clinical indices: hemoglobin (Hb), mean corpuscular volume (MCV), white blood cell (WBC) count, uric acid, and routine hemodynamic metrics. Constraining the analysis to these intersecting features establishes a standardized baseline, allowing for the evaluation of cross-population reproducibility regarding the regulatory hubs and topological characteristics identified in the primary RUC cohort.

3. Results

3.1. Baseline characteristics and methodological benchmarking

Profiling the primary derivation (RUC, $N = 7,343$) and independent validation (KNHANES, $N = 6,881$) cohorts revealed significant cross-stratum variations for most clinical parameters ($P < 0.01$) ([Table 1](#), [Table S1](#)). Progressive adiposity aligned with systemic escalations in

metabolic and inflammatory markers, notably uric acid, alanine aminotransferase (ALT), and white blood cell (WBC) counts, alongside a continuous attenuation in erythrocyte indices such as mean corpuscular volume (MCV). Demographic stratifications indicated a pronounced sex-distribution shift. Female participants constituted the majority of the normal-weight baseline, whereas male subjects predominated within the obesity strata across both datasets (74.4% in RUC, 50.3% in KNHANES). The derivation cohort also skewed younger (21.23 ± 3.59 years) relative to the validation population (47.79 ± 21.96 years). Despite these demographic heterogeneities, the conservation of macroscopic covariation patterns establishes an empirical foundation for subsequent topological modeling.

Application of conventional topological models highlighted critical limitations in resolving stage-dependent network dynamics (Fig. S1). Spearman correlation networks captured dense global covariations but could not differentiate direct dependencies from indirect associations (Fig. S1A). Gaussian Graphical Models (GGMs) improved structural sparsity but yielded undirected backbones that remained largely invariant across the four BMI stages, obscuring dynamic rewiring (Fig. S1B). Furthermore, Pseudo-Time-Lagged Cross-Correlation (TLCC) networks proved highly susceptible to cross-sectional sampling variance, generating a fragmented topology exclusively within the obesity stratum (Fig. S1C).

Deploying the ODE-based idopNetwork framework circumvented these constraints by inferring directed, weighted regulatory topologies across the distinct BMI strata (Fig. 1, top panels). Quantitative profiling of degree centralities corroborated a macroscopic architectural reorganization (Fig. 1, bottom panels). Rather than reflecting localized parameter fluctuations, progressive adiposity aligned with a fundamental transition in core regulatory hub identities. Subsampling-based stability analysis ($B = 100$; Fig. S2) mathematically substantiated the statistical robustness of this inferred systemic rewiring.

3.2. Topological reorganization of core hematometabolic hubs along the ALI gradient

Mapping the quasi-dynamic trajectories of individual parameters against the continuous Allostatic Load Index (ALI) gradient unmasked nonlinear adaptations across the physiological axes (Fig. 2, Fig. S3) [34]. Non-obese cohorts maintained an ordered allometric scaling pattern, where the majority of metabolic and inflammatory biomarkers exhibited stable, mild positive scaling ($\beta > 0$). Escalating to the obesity stratum coincided with a distinct topological phase transition. Key inflammatory and metabolic variables (WBC, neutrophils, monocytes, uric acid, and ALT) mapped steeper positive slopes across the higher ALI range, mathematically reflecting an intensified systemic response to the increasing allostatic load [4,35]. Tracking the dynamic degree centralities of core hubs (Fig. 3) demonstrated that this macroscopic reorganization was driven by fundamental shifts in node polarity and hierarchical dominance, rather than mere proportional scaling.

The macroscopic behavior of hemoglobin (Hb) provided the most prominent signature of this topological rewiring, undergoing a complete polarity inversion across the BMI spectrum. Within the underweight and normal-weight networks, Hb operated as a primary homeostatic anchor characterized by a dense matrix of directed negative (inhibitory) edges (Fig. 1A and B). This specific configuration exhibited persistent negative predictive precedence over diverse immune and inflammatory markers (such as monocytes, eosinophils, and lymphocytes). Mathematically, this structural embedding maps a macroscopic covariation that aligns with the dampening of sterile inflammation under baseline conditions [36,37]. Advancing into clinical obesity, the Hb hub underwent a profound topological reversal (Fig. 1D). Abandoning its baseline inhibitory signature, Hb projected dense positive (promoting) topological edges toward hemodynamic indices (SBP, DBP) and inflammatory cellular subsets (neutrophils, basophils). Rather than implying direct

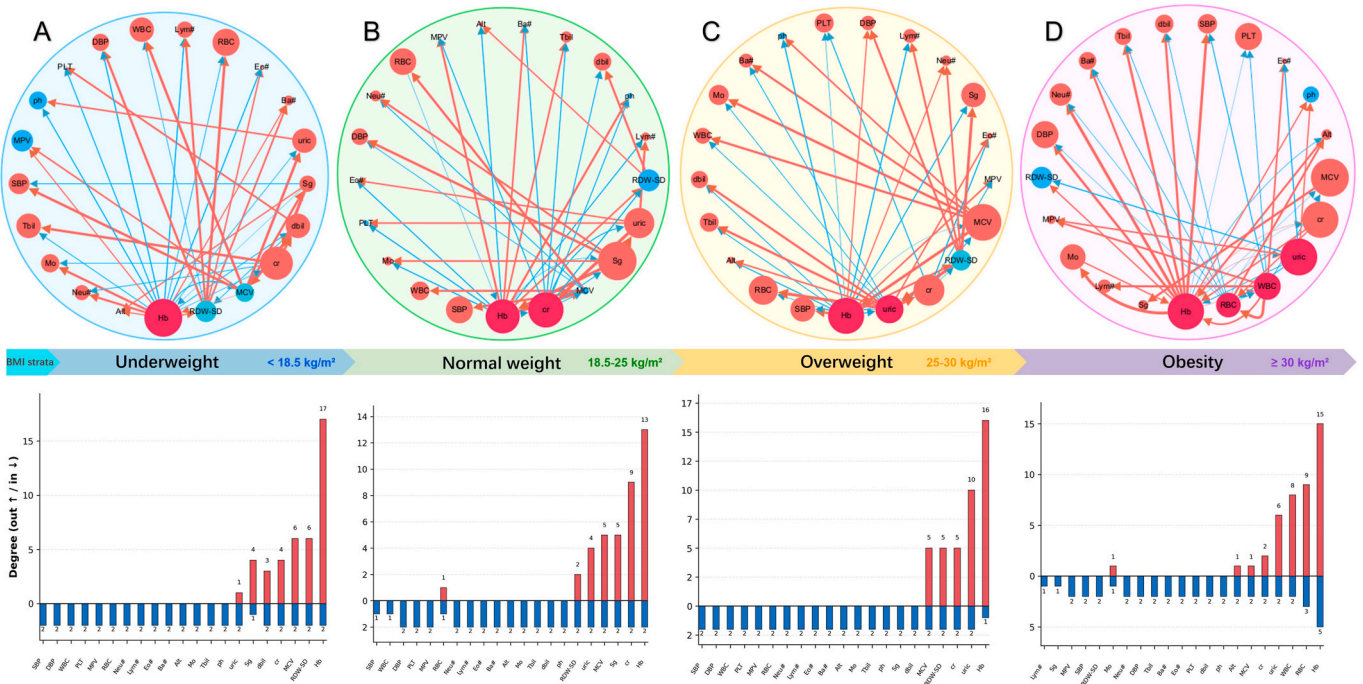


Fig. 1. Topological rewiring and quantitative degree centrality profiling of BMI-stratified physiological regulatory networks. (Top panels) Directed, signed interaction networks reconstructed via the idopNetwork framework across four BMI strata (underweight, normal weight, overweight, and obesity). Each network maps dynamic regulatory relationships among 21 clinical variables. Node size is proportional to intrinsic self-effect strength, reflecting the homeostatic capacity of each variable. Node color encodes self-regulatory direction (red: net self-activation; blue: net self-inhibition). Directed edges represent inferred directional regulatory dependencies; edge color denotes interaction sign (red: positive; blue: negative), and edge thickness corresponds to interaction magnitude. **(Bottom panels)** Back-to-back bar charts quantify topological centrality. Upward red bars represent out-degree (systemic regulatory drivers/source hubs), and downward blue bars represent in-degree (regulatory targets). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

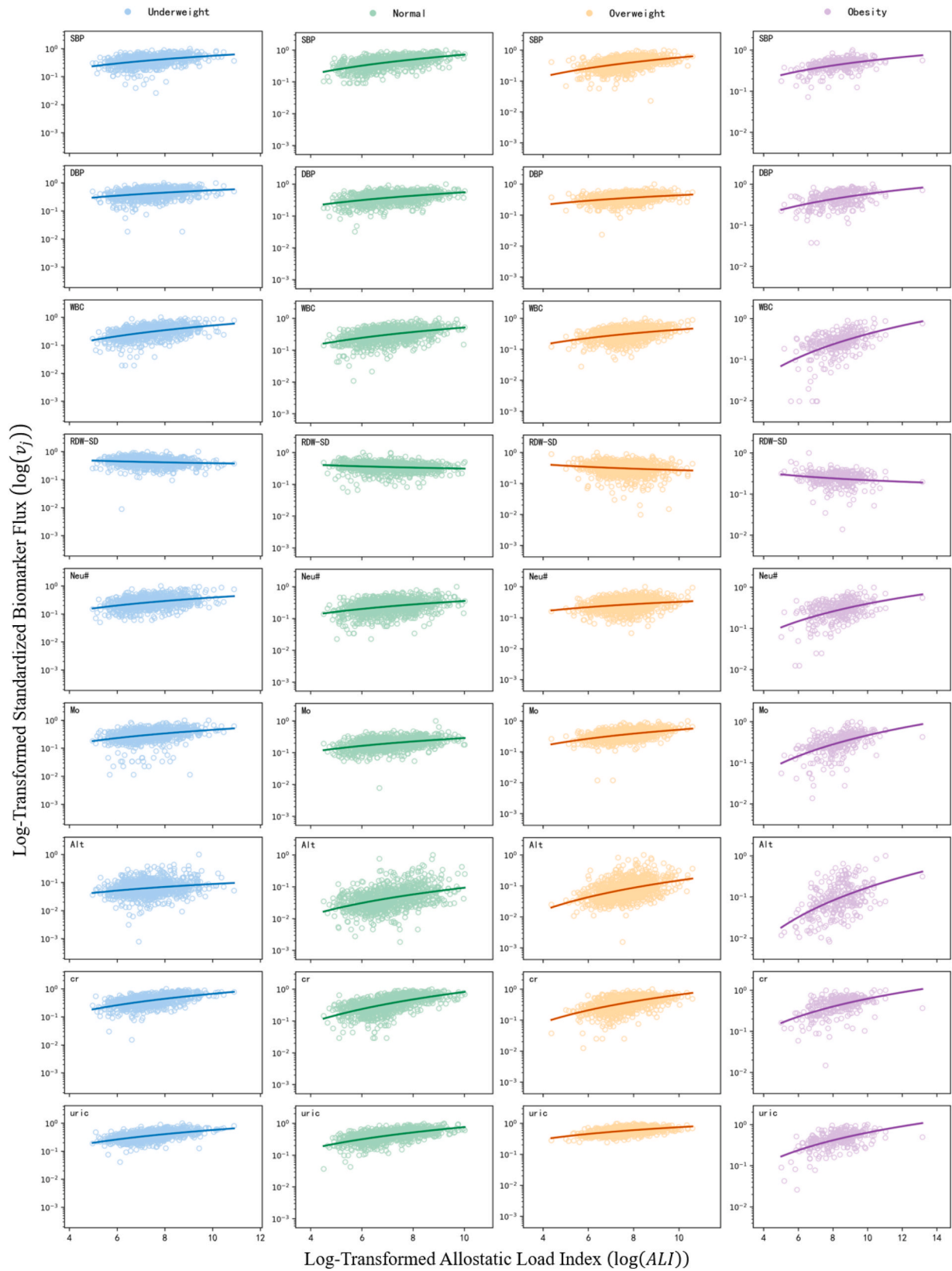


Fig. 2. Quasi-dynamic allometric scaling of hematometabolic biomarkers across BMI strata. A power-law allometric scaling function reconstructs the quasi-dynamic trajectories of nine core biomarkers (SBP, DBP, WBC, RDW-SD, Neu#, Mo, ALT, Cr, and Uric Acid) across four body mass index (BMI) strata. Individual min-max-normalized observations are plotted on a log-scale Y-axis against the continuous Allostatic Load Index (ALI) gradient on the X-axis. This mathematical embedding quantifies the aggregate physiological burden from cross-sectional clinical data. Open colored circles denote individual observations, while solid curves represent the fitted stage-dependent, non-linear allometric scaling trajectories.

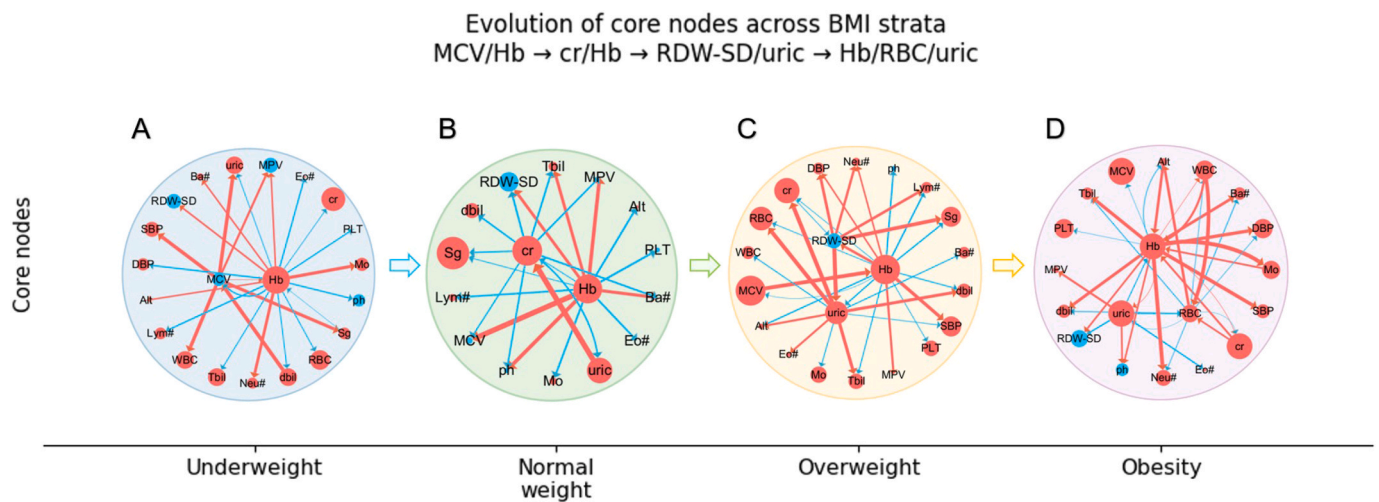


Fig. 3. Cross-stratum evolution of core regulatory hubs and associated physiological mechanisms. Dominant hub nodes transition across BMI strata (underweight, normal weight, overweight, and obesity), mapping structural changes against physiological interpretations. Hemoglobin (Hb) shifts from a homeostatic maintainer in leaner states to a pathological driver in obesity, aligning with hemodynamic stress. Uric Acid and RDW-SD emerge as critical bottleneck nodes in the overweight and obesity strata, coordinating systemic metabolic rigidity and oxidative stress, respectively.

biochemical catalysis, this altered predictive topology captures the systemic macroscopic convergence of hypoxia-associated erythropoiesis with exacerbated cardiovascular stress [38–40].

Operating in tandem with the Hb inversion, the trajectory of uric acid mapped a highly biphasic structural transition. Throughout the leaner baseline stages, uric acid remained a downstream, buffered node. The transition into the overweight stratum, however, precipitated its

rapid centralization into a dominant topological hub (Fig. 1C). Decoupling these variable-specific effects (Fig. 4) revealed that uric acid generated robust positive predictive coupling with intrinsic erythrocyte lineages, creatinine, and specific hepatic enzymes (ALT, TBIL), mathematically elevating the dynamic trajectories of these target nodes above their intrinsic baselines [41]. The subsampling-based stability analysis confirmed uric acid as a principal systemic driver in the overweight

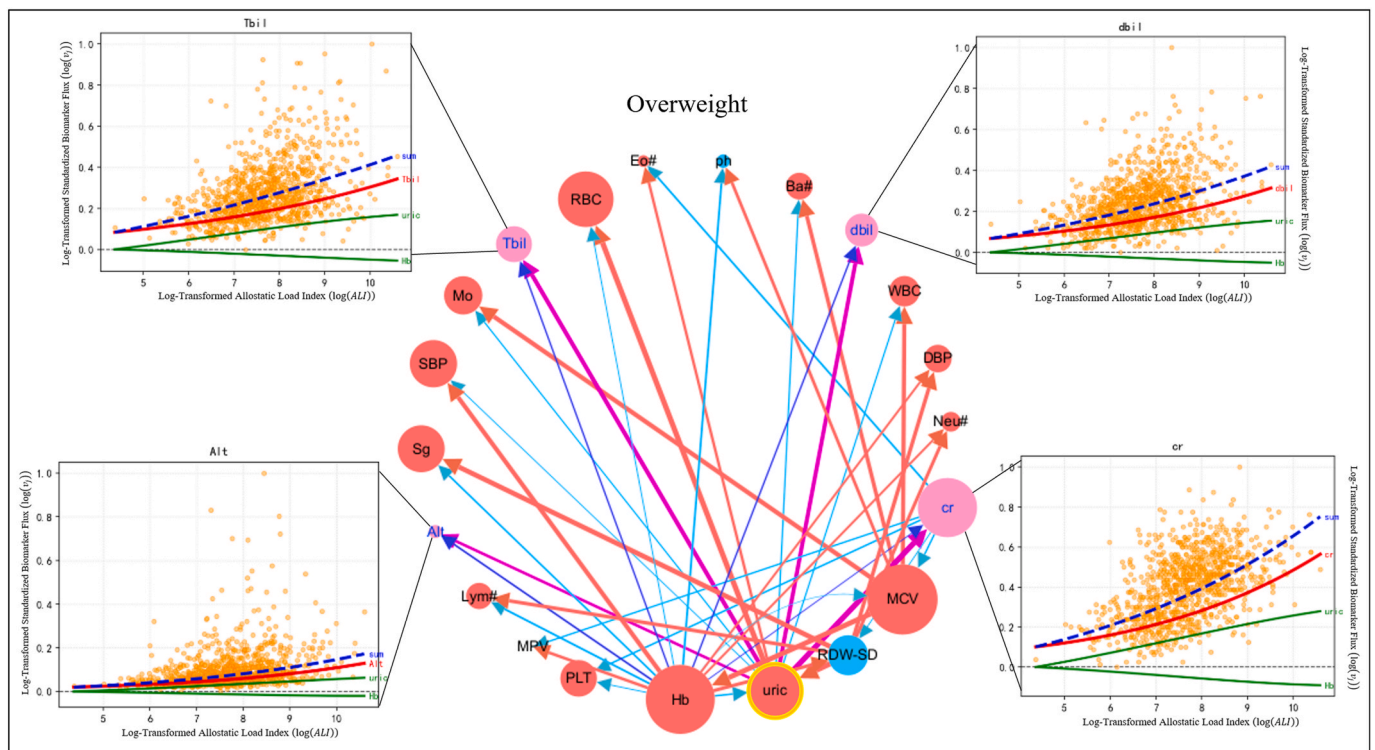


Fig. 4. Multi-scale visualization of Uric Acid-driven metabolic dysregulation in the overweight network. (Center) Directed idopNetwork. Node size and edge thickness reflect intrinsic homeostatic capacity and interaction magnitude, respectively (red/blue edges: positive/negative effects). Highlighted pathways (purple and blue arrows) delineate targeted regulatory pressures from Uric Acid and Hemoglobin (Hb) onto key hepato-renal markers (Cr, TBIL, DBIL, ALT). **(Peripheral)** Dynamic decoupling plots for these targets along the *ALI* gradient. Positive topological coupling from Uric Acid (upper green lines) structurally elevates the total dynamic trajectory (blue dashed lines) of target nodes above their intrinsic baselines (red solid lines). Concurrent inhibitory vectors from Hb (lower green lines) accompany this elevation, capturing the mechanistic centralization of Uric Acid in exacerbating systemic metabolic burden. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

cohort. Progressing into the obese state degraded this transient hierarchical dominance. Uric acid transitioned into a highly constrained downstream node, emitting concentrated negative topological edges that topologically correspond to localized structural blockades and diminished metabolic buffering capacity [42].

Isolating the dynamic topology of Red Cell Distribution Width (RDW-SD) further delineated the macroscopic transition from functional elasticity to interactive exhaustion (Fig. 5, Fig. S4). Across non-obese states, the total dynamic effect of RDW-SD demonstrated adaptive fluctuations, remaining close to or mildly above its intrinsic self-regulatory baseline through transient positive topological reinforcement from Hb. The escalating systemic load in the overweight and obese phases precipitated a sharp downward crossover of RDW-SD below its intrinsic baseline. Mathematically, this trajectory shift aligned with waning positive coupling from Hb and escalating negative topological suppression from creatinine. Subjected to biphasic perturbations and alternating directional influences in the advanced adiposity stage, the RDW-SD trajectory visually and algebraically maps the exhaustion of localized hematopoietic buffering [43], signaling a broader reduction in macroscopic network resilience.

3.3. Quantifying systemic deadlock via path homology

Application of the GLMY path homology framework [28,29] enabled the quantification of higher-order structural complexity and directed information flow within the physiological networks. Persistent Betti numbers ($\beta_0, \beta_1, \beta_2$) derived through directed network filtrations generated distinct topological barcodes across the BMI spectrum (Fig. 6). The normal-weight baseline architecture organized primarily around multiple persistent β_1 cycles (Fig. 6B). These unconstrained one-dimensional topological cycles function as mathematical proxies for dynamic homeostatic feedback loops (such as creatinine-mediated buffering), providing the physiological elasticity essential for systemic stability.

Escalating allostatic load in the overweight stratum coincided with a pronounced topological depletion (Fig. 6C). The significant reduction of previously stable β_1 cycles provides macroscopic evidence of early systemic degradation. Rather than implying a localized micro-biochemical breakdown, this structural simplification aligns with the centralization

of metabolic hubs such as uric acid. The mathematical degradation of basal negative-feedback architectures indicates a linearization of regulatory pathways, which effectively reduces the dynamic buffering capacity of the overarching system.

The obese state exhibited an accumulation of long-lived β_1 cycles alongside the emergence of multiple persistent β_2 voids, delineating a severe loss of structural flexibility (Fig. 6D). Prolonged β_1 features in this advanced stage suggest a pathological constraint where elevated parameters, including the hypoxia-associated Hb trajectory, become topologically trapped within rigidified feedback loops. Concurrently, unlike the singular, localized structural constraint observed in the underweight group, the multi-dimensional generation of β_2 voids in advanced adiposity mathematically models extensive pathway blockades. These higher-order algebraic signatures indicate systemic pathway desensitization rather than transient parameter fluctuations. Collectively, the simultaneous accumulation of inflexible cycles and topological voids quantifies a severely compromised internal milieu, mathematically delineating the macroscopic transition toward an allostatic deadlock [44].

3.4. Sex-specific topological dimorphism in adiposity

Sex-stratified network analyses revealed that the generalized topological rigidity in the unstratified obese state masks a distinct macroscopic dimorphism. The aggregate structural constraints characterizing overall adiposity represent a mathematical superposition of two divergent allostatic trajectories.

Within the male cohort, the transition to obesity coincided with a pronounced accumulation of one-dimensional cyclic loops (β_1) (Fig. 7A). This specific architecture structurally aligned with the network centralization of myeloid immune components, particularly monocytes and total leukocyte (WBC) populations, which projected dense positive predictive edges across the internal milieu. Unresolved inflammatory parameters appeared structurally integrated within rigid positive feedback cycles, rather than indicating transient biomarker fluctuations. These persistent β_1 features mathematically delineate a “hyper-reactive” topological state, providing algebraic coordinates that correspond to the chronic metaflammation classically associated with male adiposity [45].

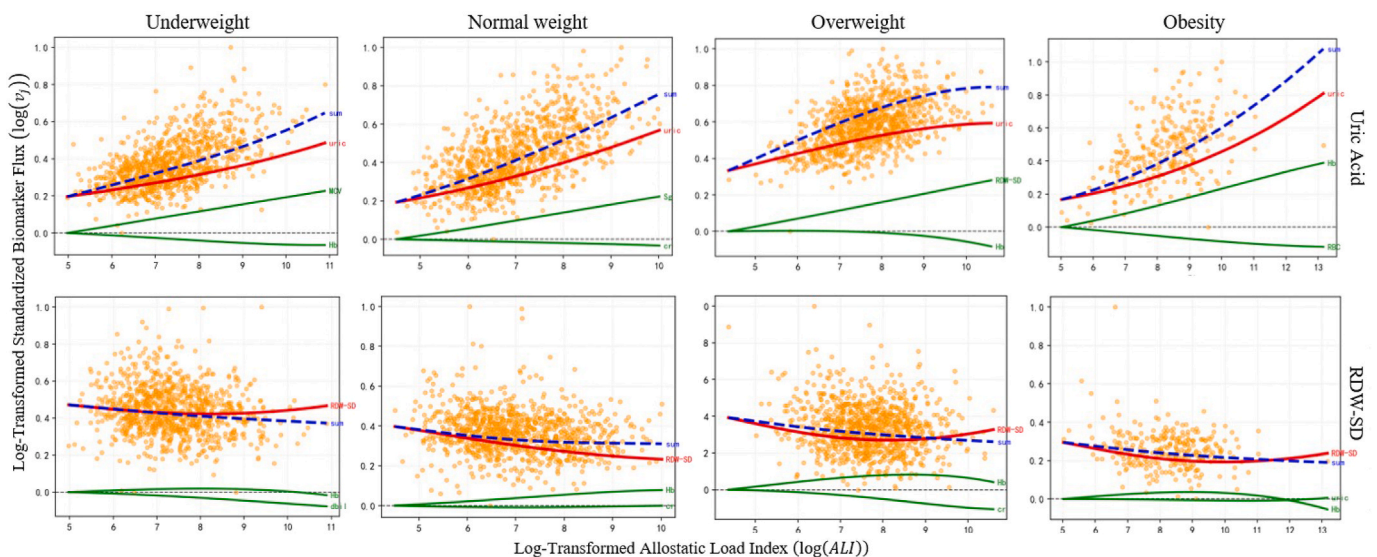


Fig. 5. Decomposition of dynamic regulatory effects for Uric Acid and RDW-SD across BMI strata. Regulatory dynamics governing Uric Acid (top) and RDW-SD (bottom) are decoupled along the continuous gradient of the log-transformed Allostatic Load Index (ALI). For each stage, the total dynamic effect (blue dashed line) of the biomarker is decomposed into its intrinsic self-regulatory contribution (red solid line) and external interactive contributions from the systemic network (green solid lines, with specific effectors such as Hb or RBC denoted). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

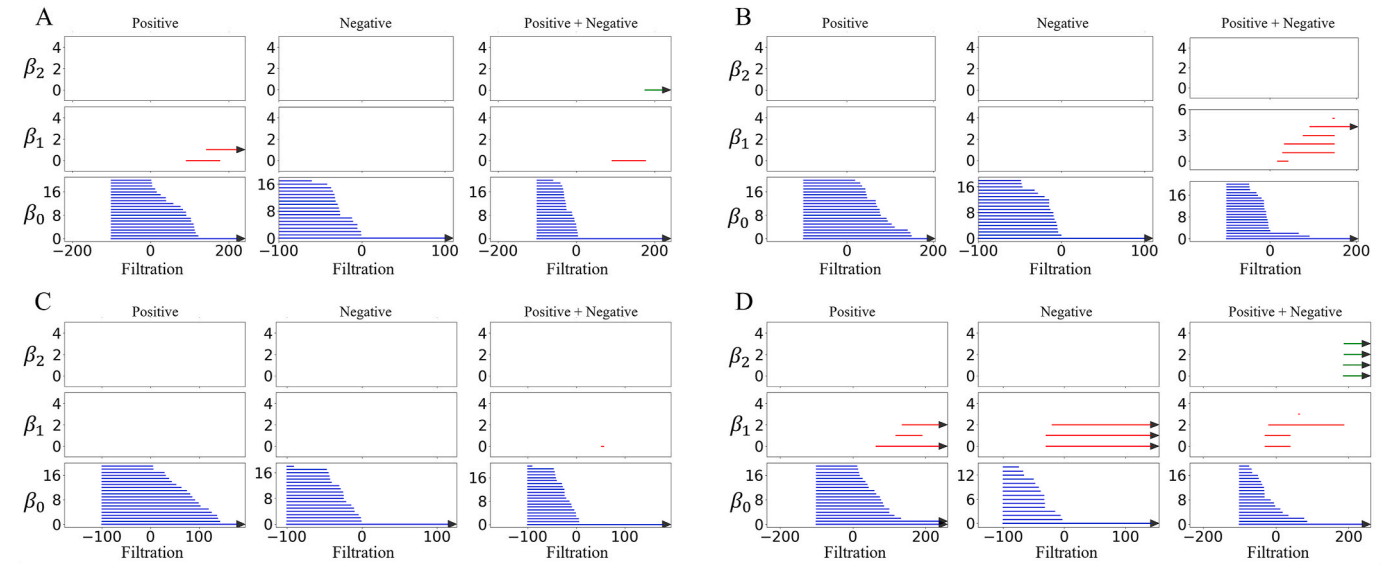


Fig. 6. GLMY path homology analysis of physiological regulatory networks. Persistent Betti barcodes computed over directed network filtrations for four groups: (A) Underweight, (B) Normal weight, (C) Overweight, and (D) Obesity. The Y-axis represents the ranks of homology groups at dimension zero (β_0 , connected components), one (β_1 , directed cyclic feedback loops), and two (β_2 , higher-order topological voids). The X-axis denotes the filtration threshold (interaction strength limits). The analysis evaluates this threshold for positive interactions (promotion, left column), negative interactions (inhibition, middle column), and the combined complete network (right column).

The female cohort exhibited an independent structural trajectory characterized by extensive topological depletion (Fig. 7B). Homeostatic β_1 cycles remained stable across leaner baselines; however, the female network demonstrated a cyclic reduction during the overweight phase, accompanied by the emergence of two-dimensional β_2 voids. This localized multi-dimensional occlusion coincided with the targeted topological expansion of the hemoglobin (Hb) hub, indicating a systemic architectural shift associated with hypoxia-driven compensatory stress. In advanced clinical obesity, higher-order homological features (β_1 and β_2) diminished almost entirely, even as the baseline capacities of Hb and WBC were sustained. This structural impoverishment mathematically defines an “exhaustive” topological state, capturing how chronic hematopoietic compensation correlates with the attenuation of feedback buffering essential for systemic elasticity.

The generalized rigidity observed in aggregate obesity functionally obscures a sexually dimorphic underlying architecture. Evaluated through these algebraic signatures, male adiposity structurally aligns with rigid inflammatory cyclic networks, whereas female adiposity pathologically degrades into exhaustion-mediated topological depletion [46].

3.5. Cross-cohort validation of topological reorganization

Construction of a 9-variable validation sub-network using the Korea National Health and Nutrition Examination Survey (KNHANES) dataset addressed the absence of specific micro-hematological parameters in the public repository. This condensed architecture provided an independent baseline to evaluate the reproducibility of the inferred topological features (Fig. 8).

Cross-population comparisons revealed consistent macroscopic dynamics within the sub-network. Both the primary and validation cohorts featured hemoglobin (Hb) and mean corpuscular volume (MCV) as central upstream topological nodes. During the overweight stage, the KNHANES dataset exhibited a distinct architectural shift aligned with the centralization of uric acid. This metabolic node generated positive directed coupling, reflecting a transient hierarchical dominance rather than isolated biomarker fluctuations. The trajectory of uric acid then degraded into a downstream node within the advanced obesity stage. Such phase evolution directly parallels the structural rewiring mapped

in the primary RUC network.

The polarity shift of Hb provided a primary metric for validation. In the non-obese stages of the primary network, Hb predominantly generated negative predictive vectors. The external cohort mirrored this inhibitory pattern across the normal-weight and overweight phases. At the obesity stage, the primary architecture displayed a polarity inversion of the Hb node, marked by the emergence of dense positive directed edges. The external validation network replicated this structural shift, capturing the specific emergence of positive promoting edges from the altered Hb hub. Although the overall topological intensity appeared slightly attenuated in the independent dataset, the cross-population analysis confirms the macroscopic transition of Hb toward positive topological regulation under extreme adiposity.

4. Discussion

The idopNetwork framework successfully reconstructed directed, cyclic hematometabolic regulatory networks by embedding cross-sectional clinical observations into quasi-dynamic trajectories. Decoupling directed regulatory flows from systemic confounding bypasses the limitations inherent to conventional linear baselines. The inferred macroscopic architecture demonstrates that progressive adiposity extends far beyond quantitative fluctuations in isolated biomarkers. Instead, this reorganization dictates a fundamental phase shift in systemic mathematical regulation, tracing the macroscopic transition from physiological homeostasis to an adiposity-induced allostatic deadlock.

At the core of this systemic reorganization, the hematopoietic axis undergoes a complete functional inversion. Within leaner states, a stable hematological-renal backbone anchors the internal milieu. The central hemoglobin (Hb) node acts as a suppressive hub, projecting dense inhibitory vectors that mathematically correlate with dampened baseline immune activation. Simultaneously, its trajectory establishes a strong predictive coupling with RDW-SD, structurally aligning with the physiological preservation of vital oxygenation. Advancing into the obese stratum precipitates a profound topological reversal. Transforming into a global promoter, the Hb hub projects positive predictive edges toward distinct hemodynamic and inflammatory variables. This altered signature captures a severe maladaptive response to obesity-associated tissue microhypoxia, indicating that sustained allostatic

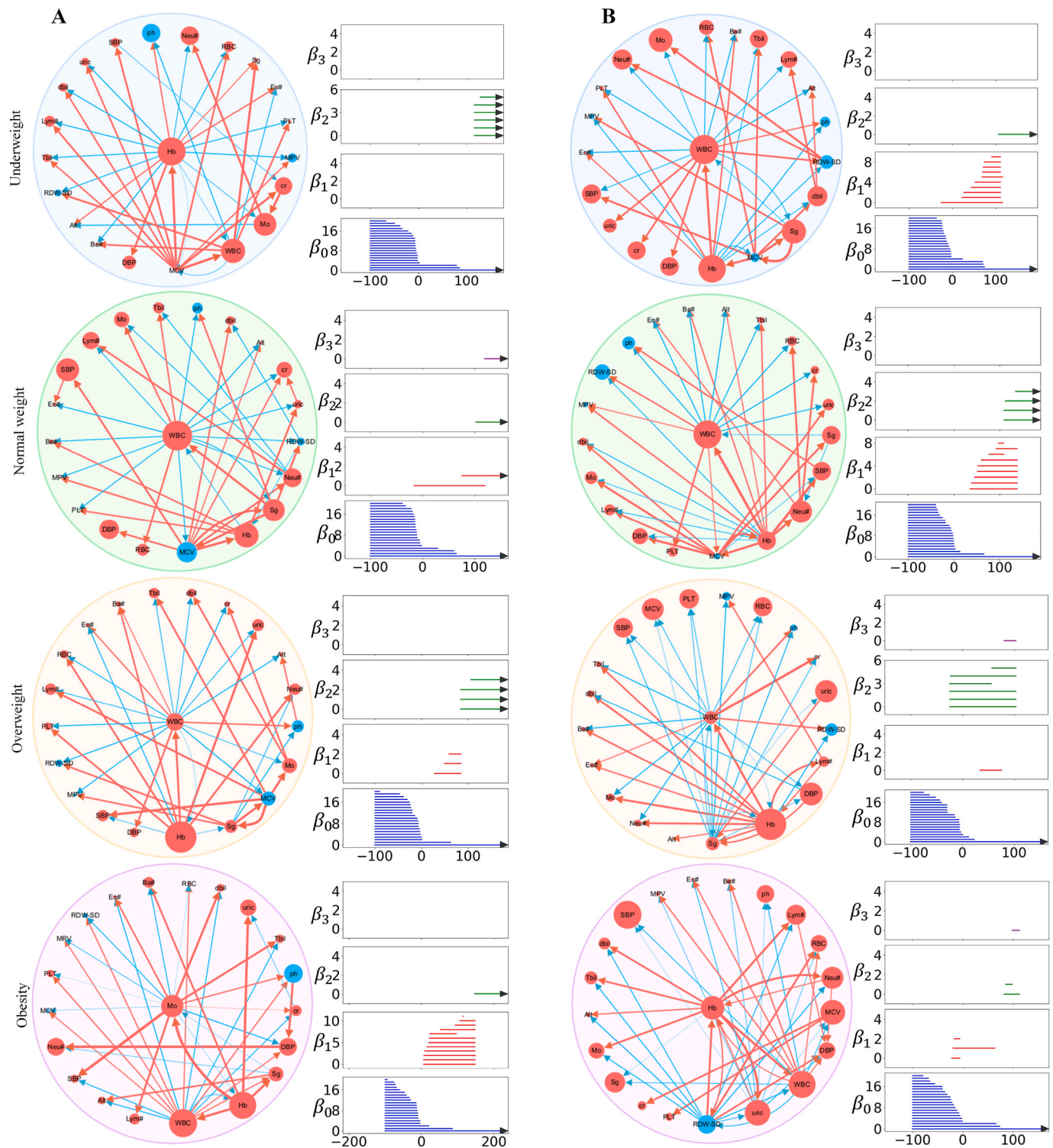


Fig. 7. Sex-specific topological dimorphism and divergent deadlock trajectories. (A) Male and (B) Female cohorts. Left panels: directed idopNetworks reconstructed across four BMI strata. Node size reflects intrinsic homeostatic capacity; red/blue nodes indicate net self-activation/inhibition; red/blue edges denote directional regulatory promotion/inhibition. Right panels: corresponding GLMY persistent Betti barcodes, tracking connected components (β_0 , blue), 1-dimensional cyclic loops (β_1 , red), 2-dimensional topological voids (β_2 , green), and 3-dimensional structures (β_3 , purple). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

stress forces early compensatory mechanisms to pivot and structurally reinforce pathogenic cascades [39,40].

Trajectory analysis identifies the overweight stratum as the critical physiological tipping point for this structural collapse. The macroscopic network departs from stable homeostatic negative feedback loops well

before the onset of clinical obesity, driven by the rapid centralization of uric acid. Rather than reflecting isolated metabolic accumulation, this node establishes robust positive topological coupling with hepatic enzymes and erythrocyte indices. Such systemic rewiring supplies mathematical evidence for the severe peripheral organ burden associated

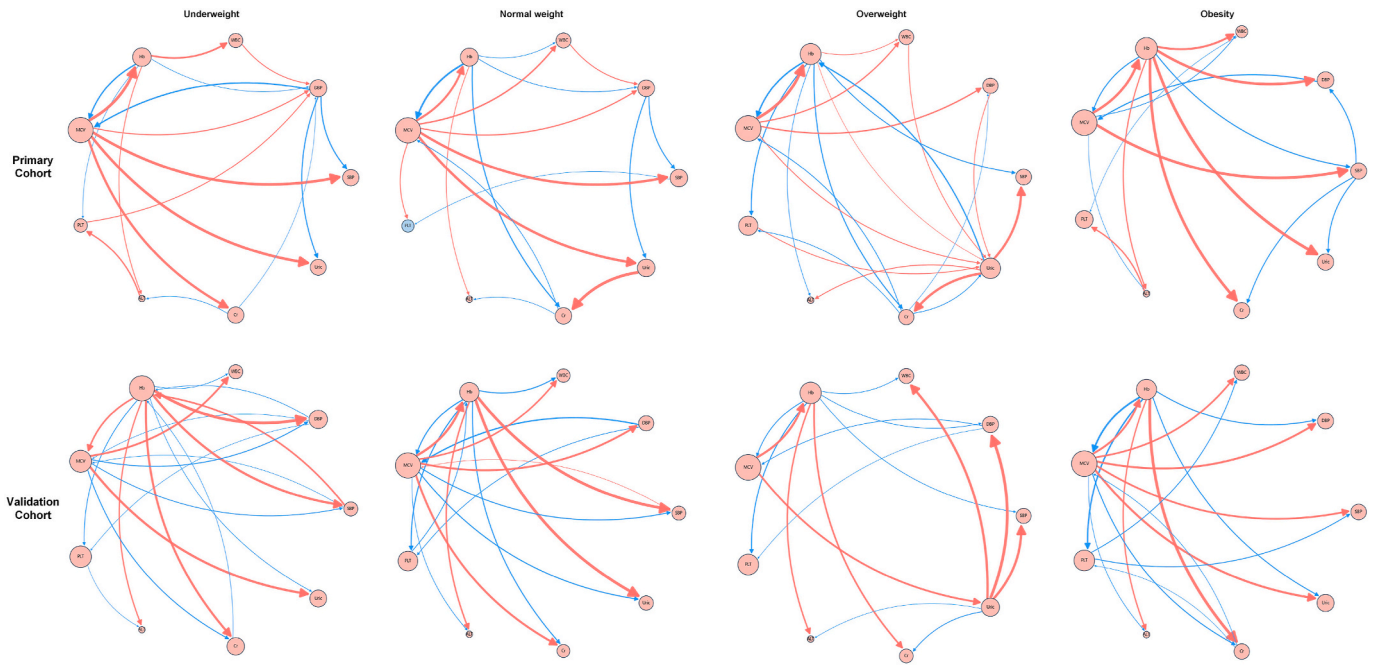


Fig. 8. Cross-cohort topological validation of metabolic networks. (Top row) The 9-variable sub-network derived from the Primary Cohort (RUC), preserving core variables. **(Bottom row)** The external validation network constructed using the KNHANES dataset. Red arrows represent positive regulation; blue arrows represent negative regulation. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

with early hyperuricemia [41]. Operating as a topological bridge within this altered architecture, RDW-SD structurally links localized erythrocyte stress with broad systemic metabolic dysregulation. The subsequent collapse of its allometric scaling directly maps the terminal exhaustion of intrinsic hematopoietic buffering capacity, finalizing the breakdown of the overarching physiological network.

Application of GLMY path homology provides the rigorous algebraic basis for quantifying this macroscopic structural collapse [28,29]. Within the normal-weight baseline, persistent one-dimensional cyclic loops (β_1) serve as the primary mechanism for dynamic buffering, supplying essential network elasticity. The obese state disrupts this homeostatic configuration, triggering an abnormal accumulation of maladaptive β_1 loops alongside the concurrent emergence of two-dimensional topological voids (β_2). The proliferation of these persistent β_1 loops mathematically maps to unconstrained, hyperactive systemic feedback cycles rather than direct physical molecular entanglement. Subsequent generation of β_2 voids structurally aligns with localized pathway blockades. Collectively, these precise algebraic signatures denote a severely compromised internal milieu, demonstrating how advanced adiposity restricts overall network flexibility and degrades systemic resilience [47].

Sex-stratified analyses explicitly dissect the heterogeneity of this macroscopic gridlock, revealing that the overarching aggregate network represents a mathematical superposition of two fundamentally divergent pathogenic trajectories. The male cohort demonstrates a preferential accumulation of inflammatory β_1 cyclic loops aligned with the centralization of myeloid immune cells. This architecture defines a “hyper-reactive” deadlock, capturing the rigid positive feedback cycles that sustain the chronic metaflammation classically associated with male adiposity [45,48,49]. The female cohort traces an independent pathological route culminating in extensive topological depletion. While two-dimensional β_2 voids emerge transiently during the overweight phase as indicators of initial pathway occlusion, progression into advanced clinical obesity extinguishes these higher-order homological features almost entirely. Driven by severe structural impoverishment,

this “exhaustive” deadlock captures the complete attenuation of compensatory feedback buffering. The female internal ecosystem degrades into a profound topological barrenness, starkly contrasting with the inflammatory storms observed in males [46,49,50]. Recognizing this profound structural dimorphism fundamentally challenges universal treatment protocols. These algebraic distinctions highlight the necessity for sex-specific precision interventions—suggesting that therapeutic strategies might target the disruption of rigid inflammatory cycles in males, while prioritizing the restoration of global network elasticity and mitigation of hematopoietic exhaustion in females.

Analysis of the targeted 9-variable sub-network from the KNHANES external cohort provided independent verification of these macroscopic transitions. The external dataset replicated fundamental phase trajectories mapped in the primary derivation cohort, specifically confirming the structural centralization of uric acid during the overweight stage and the subsequent Hb-associated topological reversal within the obese stratum. Alongside these conserved architectural motifs, distinct cohort-specific heterogeneities emerged. The KNHANES obesity network exhibited an attenuated topological radiation intensity for Hb compared to the primary population. Rather than diminishing the robustness of the core findings, this structural divergence captures the inherent baseline variations between a localized clinical cohort and a broad national epidemiological survey. Accommodating wider demographic, dietary, and genetic diversity, this variance validates the analytical capacity of the idopNetwork framework to resolve population-specific evolutionary pathways without losing the fundamental signatures of systemic structural collapse.

While the combination of the idopNetwork framework with allometric scaling effectively maps macroscopic systemic reorganization, certain methodological constraints require explicit acknowledgment. Crucially, while subsampling stability and external validation confirm the internal consistency of the inferred topologies, these computational assessments do not establish biological ground truth. Moreover, the Allostatic Load Index (ALI) functions as a mathematical proxy for disease progression, inherently relying upon macroecological assumptions

governing resource-constrained systems. Although GLMY path homology captures macroscopic structural failures by identifying maladaptive β_1 loops and β_2 voids, routine hematometabolic biomarkers lack the granular resolution needed to pinpoint the intracellular signaling cascades driving these phase transitions. Bridging the gap between these macroscopic topological deadlocks and their precise molecular effectors will ultimately require targeted in vivo perturbations.

Furthermore, because the clinical inputs originate from static cross-sectional repositories, the directed network edges strictly represent predictive mathematical dependencies rather than definitive chronological causality. Because neither cohort provides longitudinal data, the prognostic value of the inferred network features for incident metabolic outcomes remains untested. While subsampling confirms internal structural reproducibility, it cannot substitute for temporal validation. Future prospective cohorts with repeated measurements are essential to determine whether these topological transitions represent genuine pathophysiological remodeling, and whether baseline hub signatures (e. g., hemoglobin polarity) can predict the onset of cardiometabolic diseases [51,52].

5. Conclusions

This investigation explicitly maps the systemic topological evolution of the human hematometabolic network across progressive BMI strata. By integrating the idopNetwork framework with GLMY path homology, the analytical pipeline captures a graded non-linear structural transition along the ALI gradient, wherein the internal physiological ecosystem structurally deteriorates from robust homeostatic buffering into an advanced allostatic deadlock. The structural centralization of uric acid defines the initial systemic response to escalating adiposity, operating not as a direct physical catalyst but as a predictive mathematical antecedent to the profound functional inversion of hemoglobin (Hb). Through this inversion, the topological signature of Hb transitions from a baseline physiological anchor to a pro-inflammatory correlate.

Advanced clinical obesity manifests as a rigid systemic gridlock, quantified by the accumulation of maladaptive β_1 cyclic loops and β_2 topological voids. This terminal architecture exhibits profound sexual dimorphism: male adiposity degrades into hyper-reactive inflammatory cyclic tangling, whereas the female trajectory culminates in exhaustive topological barrenness. These precise macroscopic algebraic coordinates challenge universal, one-size-fits-all treatment protocols, highlighting the fundamental necessity for sex-specific precision interventions in managing obesity-driven cardiometabolic risk. Longitudinal cohorts with repeated measurements are required to validate the prospective predictive value of these inferred network features.

CRedit authorship contribution statement

Jiapeng Wang: Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Yu Wang:** Supervision, Software, Project administration, Methodology. **Xin Tang:** Resources, Data curation. **Xinrui Luo:** Software. **Huixian Yan:** Resources. **Wencheng An:** Data curation. **Jiayi Li:** Methodology. **Jiayu Xu:** Formal analysis. **Rongling Wu:** Conceptualization. **Huyle Liang:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Ethics approval for the primary cohort

Prospective ethical approval for the primary cohort was formally granted by the Institutional Review Board (IRB) of the Department of Psychology, Renmin University of China (IRB File No.: 26-049; Approval Date: May 19, 2026; Expiry Date: May 19, 2027). All study procedures, including the methodological protocol and informed consent documentation, underwent comprehensive institutional review and clearance prior to study initiation. Execution of the research strictly adhered

to the approved protocol constraints. This regulatory compliance encompassed obtaining signed informed consent from all enrolled participants, maintaining secure archival of consent documentation, strictly adhering to the 10-day mandatory reporting window for any adverse events, and guaranteeing the retention of all research records for a minimum of three years post-study.

Ethics approval for the validation cohort

Acquisition of the external validation dataset relied upon the Korea National Health and Nutrition Examination Survey (KNHANES), a nationally representative cross-sectional survey administered by the Korea Disease Control and Prevention Agency (KDCA). The institutional review board of the KDCA previously approved the foundational KNHANES protocol, securing written informed consent from all participants prior to their primary enrollment. Utilizing strictly de-identified, publicly available secondary data for the current cross-cohort validation analysis, the requirement for additional institutional ethical oversight was formally waived.

Consent for publication

Not applicable.

Availability of data and materials

The datasets analyzed during the current study, along with the source code for the idopNetwork pipeline, are publicly available in the GitHub repository: https://github.com/JapWang/idopNetwork_BMI.

Ethics approval statement

Conditions of Approval.

- 1. Informed Consent:** All participants must sign an IRB-approved informed consent form bearing the official IRB stamp. Signed consent forms must be properly archived.
- 2. Adverse Events:** Any adverse or unexpected events related to this study must be reported to the IRB within 10 days of occurrence.
- 3. Record Retention:** Accurate and detailed research records must be maintained for at least 3 years following study completion.
- 4. Protocol Modifications:** No significant changes to study-related materials—including research personnel, informed consent forms, study protocols, or procedures—may be made without prior IRB re-approval.
- 5. Lapse of Approval:** Should IRB approval lapse due to expiration, all research activities must cease immediately and may only resume upon re-approval.
- 6. Annual Renewal:** This approval expires automatically on the above expiration date. A new application must be submitted to continue research beyond this period.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.combiomed.2026.111910>.

List of abbreviations

Definition of hematometabolic biomarkers (0–20) used in idopNetwork analysis.

Variable ID	Definition	Abbreviation
0	Systolic Blood Pressure	SBP
1	Diastolic Blood Pressure	DBP
2	White Blood Cell count	WBC
3	Hemoglobin	Hb
4	Mean Corpuscular Volume	MCV
5	Platelet count	PLT
6	Mean Platelet Volume	MPV
7	Red cell Distribution Width-Standard Deviation	RDW-SD
8	Absolute Lymphocyte count	Lym#
9	Absolute Neutrophil count	Neu#
10	Red Blood Cell count	RBC
11	Absolute Monocyte count	Mo
12	Absolute Eosinophil count	Eo#
13	Absolute Basophil count	Ba#
14	Alanine Aminotransferase	ALT
15	Total Bilirubin	TBIL
16	Direct Bilirubin	DBIL
17	Specific Gravity	SG
18	Potential of Hydrogen	pH
19	Creatinine	Cr
20	Uric Acid	Uric Acid (UA)

Note: All biomarkers were derived from routine health examination data and were standardized prior to idopNetwork reconstruction.

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